

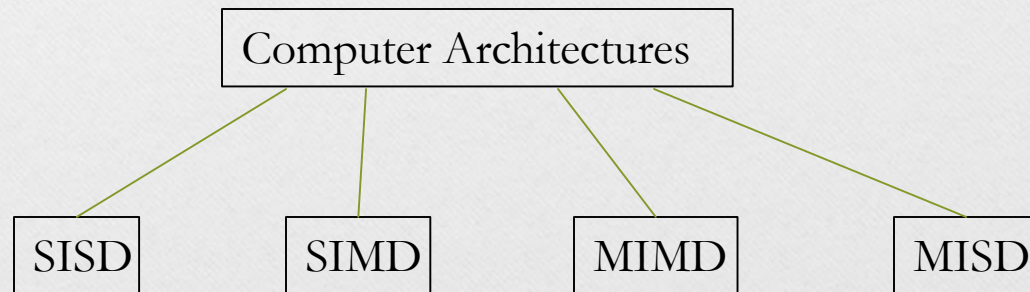
Flynn's Taxonomy

Classification of Models

- Flynn's taxonomy
- Classification based on the memory arrangement
- Classification based on communication
- Classification based on the kind of parallelism
 - Data-parallel
 - Function-parallel

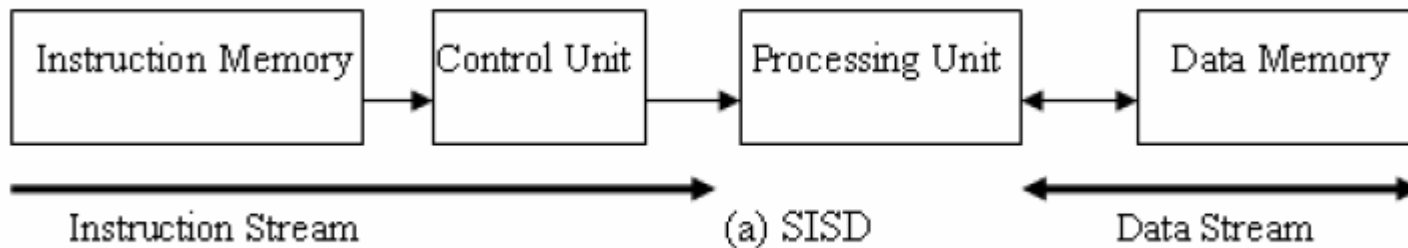
Flynn's taxonomy

- Single Instruction Single Data (SISD)
 - Traditional sequential computing systems
- Single Instruction Multiple Data (SIMD)
- Multiple Instructions Multiple Data (MIMD)
- Multiple Instructions Single Data (MISD)



SISD

- At one time, one instruction operates on one data
- Traditional sequential architecture



Instruction Stream:

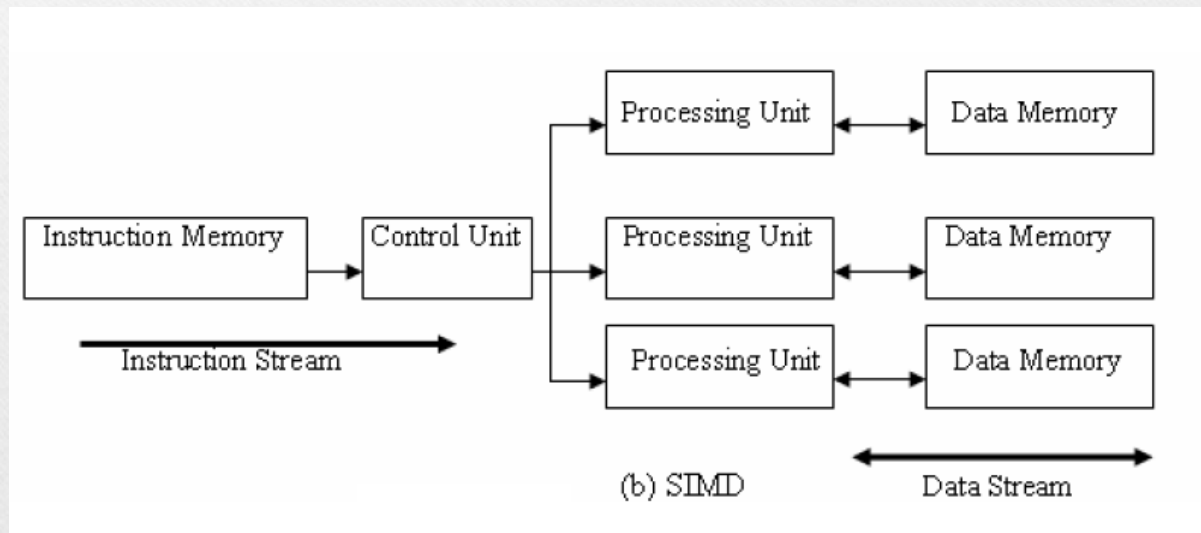
The sequence of instructions can be read from memory constitutes as instruction stream.

Data Stream:

The operations performed on the data in the processor constitutes an data stream.

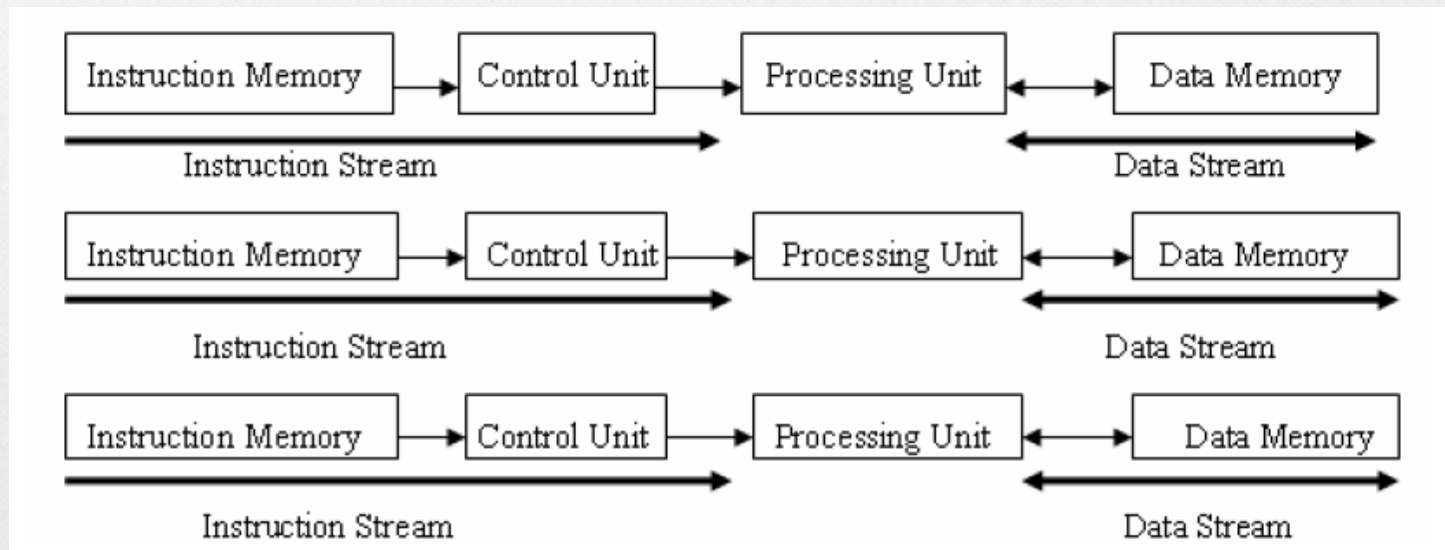
SIMD

- At one time, one instruction operates on many data
 - Data parallel architecture
 - Vector architecture has similar characteristics, but achieve the parallelism with pipelining.
- Array processors



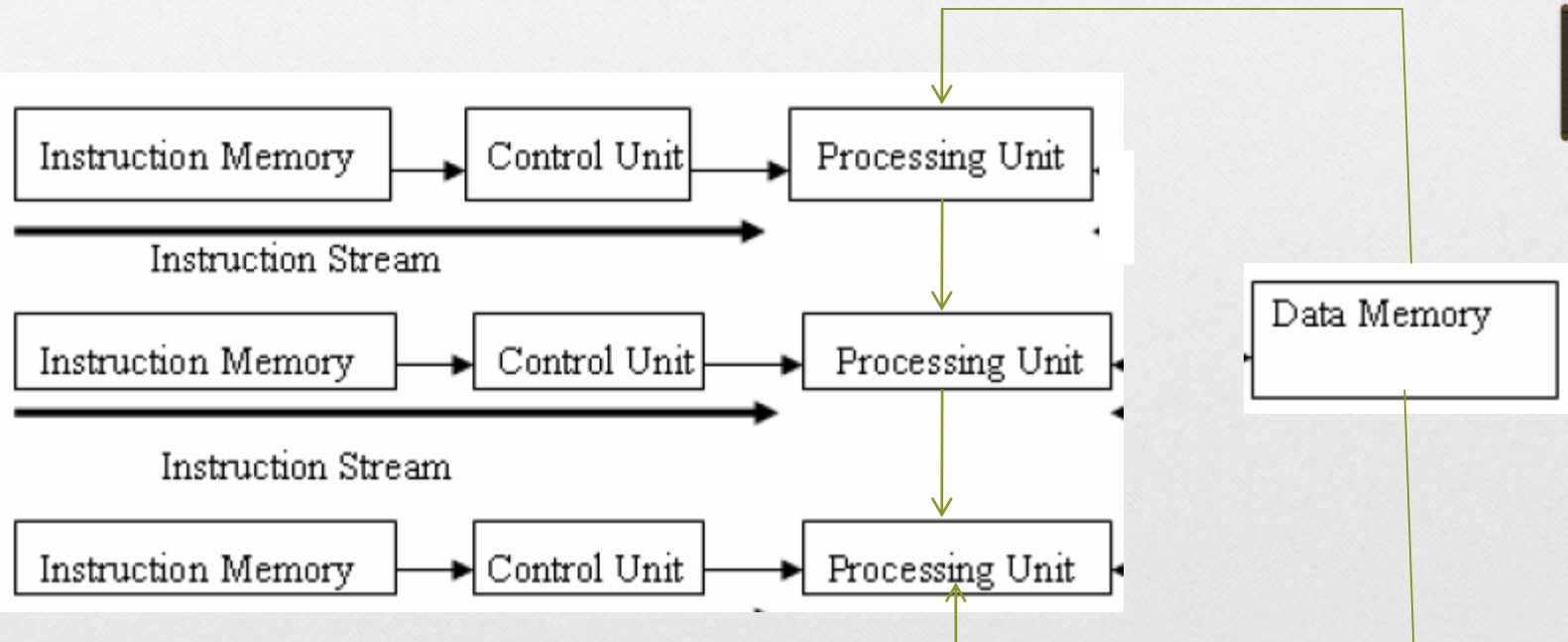
MIMD

- Multiple instruction streams operating on multiple data streams
 - Classical distributed memory or SMP architectures



MISD machine

- Not commonly seen.
- Systolic array is one example of an MISD architecture.



Flynn's taxonomy summary

- SISD: traditional sequential architecture
- SIMD: processor arrays, vector processor
 - Parallel computing on a budget – reduced control unit cost
 - Many early supercomputers
- MIMD: most general purpose parallel computer today
 - Clusters, Multicore
- MISD: not a general purpose architecture.

Advantages and Disadvantages of Flynn taxonomy

- Advantages of Flynn

- » Universally accepted
- » Compact Notation
- » Easy to classify a system.

- Disadvantages of Flynn

- » Very coarse-grain differentiation among machine systems
- » Comparison of different systems is limited
- » Interconnections, I/O, memory not considered in the scheme